

Q] The sum of the possible value of x in the equation

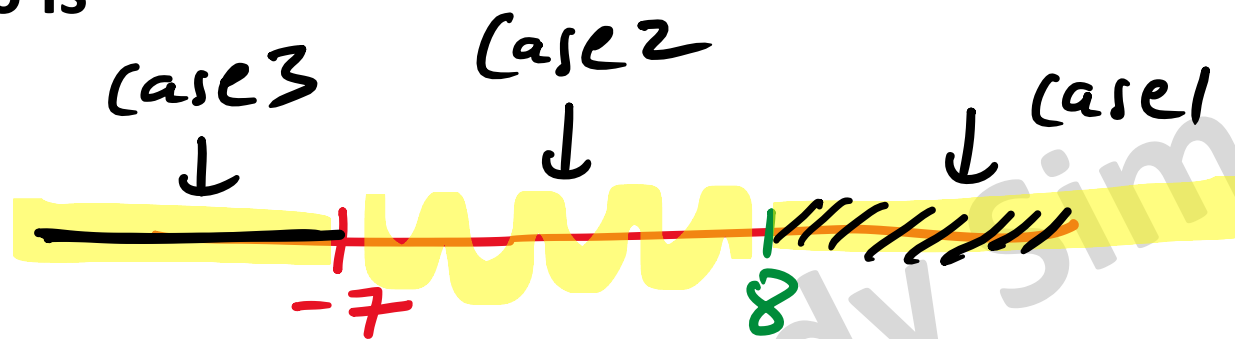
$|x+7| + |x-8| = 16$ is

(A) 0

(B) 1

(C) 2

(D) 3



CASE 1 $\rightarrow \underline{\underline{x > 8}}$:

$$(x+7) + (x-8) = 16$$

$$2x - 1 = 16$$

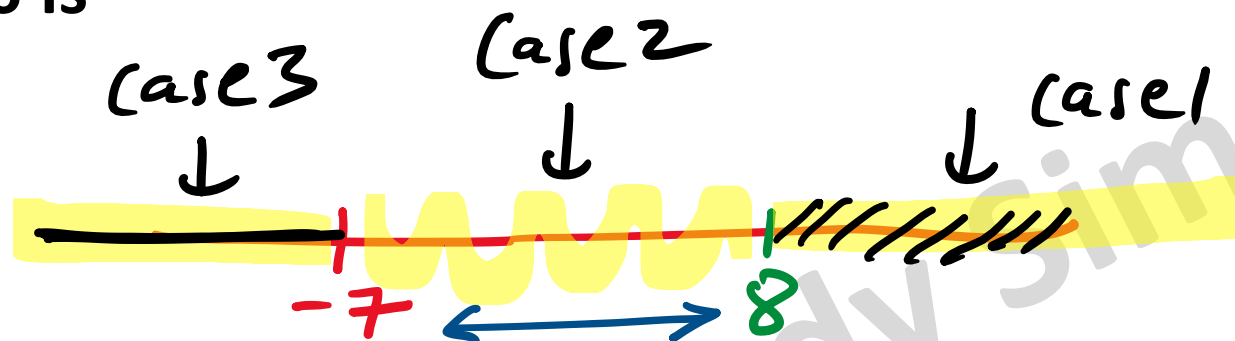
$$2x = 17$$

$$x = 8.5 //$$

Q] The sum of the possible value of x in the equation

$|x+7| + |x-8| = 16$ is

- (A) 0
- (B) 1
- (C) 2
- (D) 3



CASE 2 $\rightarrow$ $-7 < x < 8$ {Invalid case}

$$(x+7) - (x-8) = 16$$

$$\cancel{x} + 7 - \cancel{x} + 8 = 16$$

$$15 = 16 \quad [\text{Not possible}]$$

Q] The sum of the possible value of x in the equation

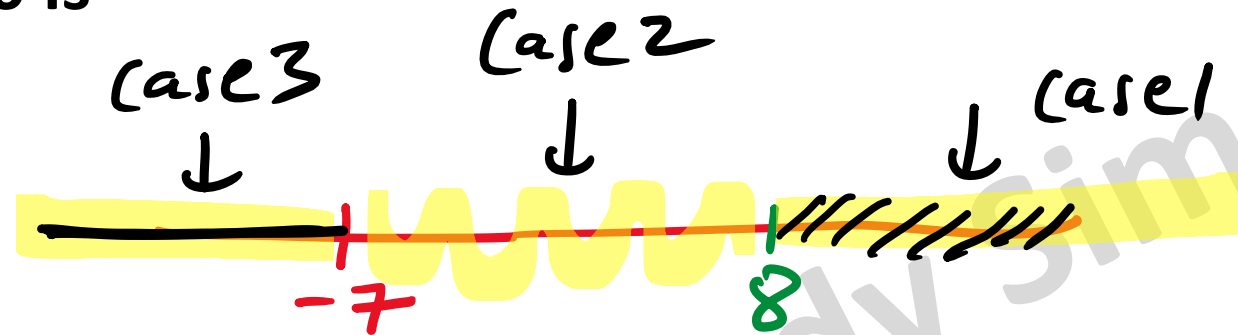
$|x+7| + |x-8| = 16$ is

(A) 0

(B) 1

(C) 2

(D) 3



CASE 3 →

$x < -7$:

$$-(x+7) - (x-8) = 16$$

$$-x - 7 - x + 8 = 16$$

$\therefore x = -7.5$

· $Ans = 8.5 - 7.5 = 1$ [option B]

Shrenik Jain — Study Simplified

Q] Function $F(a,b)$ and $G(a,b)$ are defined as follows

$F(a,b) = (a - b)^2$ and $G(a,b) = |a - b|$ where $|x|$ represents the absolute values of x . What would be the value of

$G(F(1, 3), G(1, 3))$?

(A) 2

(B) 4

(C) 6

(D) 36

[GATE EE 2018]

$$\textcircled{1} \quad F(1, 3) = (1 - 3)^2 = (-2)^2 = 4 //$$

$$G(1, 3) = |1 - 3| = |-2| = 2 //$$

$$\textcircled{2} \quad G[4, 2]$$

$$= |4 - 2|$$

$$= |2|$$

$$= 2 //$$

Q] For integer a,b,c what would be the minimum and maximum values respectively of $a+b+c$ if $\log|a| + \log|b| + \log|c| = 0$?

☒ (A) -3,3

☒ (B) -1,1

☒ (C) -1,3

☒ (D) 1,3

[GATE ME 2018]

$$\# \log P + \log Q + \log R = \log (PQR)$$

$$\log [10^0] = 0$$

$$\Rightarrow |a| |b| |c| = 10^0 = 1$$

$\Rightarrow$ If $a, b, c \rightarrow$ Integers

(Possible values)

$$\boxed{a = b = c = \pm 1}$$

$a+b+c = ?$

(Max) $\rightarrow [a=b=c=1] \rightarrow a+b+c=3 //$

(Min) $\rightarrow [a=b=c=-1] \rightarrow a+b+c=-3 //$

Q] For non-negative integer a, b, c what would be the values of $a+b+c$ if $\log a + \log b + \log c = 0$?

- ✓ (A) 3
- (B) 1
- (C) 0
- (D) -1

[GATE CE 2018]

$$\log (abc) = 0$$

$$abc = 1$$

$$a = b = c = 1$$

↓

$$a + b + c = 3 //$$

Let S_1 be the plane figure consisting of the points (x, y) given by the inequalities $|x - 1| \leq 2$ and $|y + 2| \leq 3$. Let S_2 be the plane figure given by the inequalities $x - y \geq -2$, $y \geq 1$, and $x \leq 3$. Let S be the union of S_1 and S_2 . The area of S is

(a) 26

(b) 28

☒ (c) 32

(d) 34

[GATE-2017]

$$A(S_1) = 24$$

$$+ \frac{A(S_2) = 8}{\quad}$$

$$A(S) = \underline{\underline{32}}$$

∴ ① S₁:

$$|x-1| \leq 2$$

$x > 1$: $(x-1) \leq 2$

$$x \leq 3$$

$x < 1$: $-(x-1) \leq 2$

$$-x+1 \leq 2$$

$$-x \leq 1$$

$$x \geq -1$$

$$-1 \leq x \leq 3$$

$$|y+2| \leq 3$$

$y > -2$:

$$(y+2) \leq 3$$

$$y \leq 3-2$$

$$\boxed{y \leq 1}$$

$y < -2$:

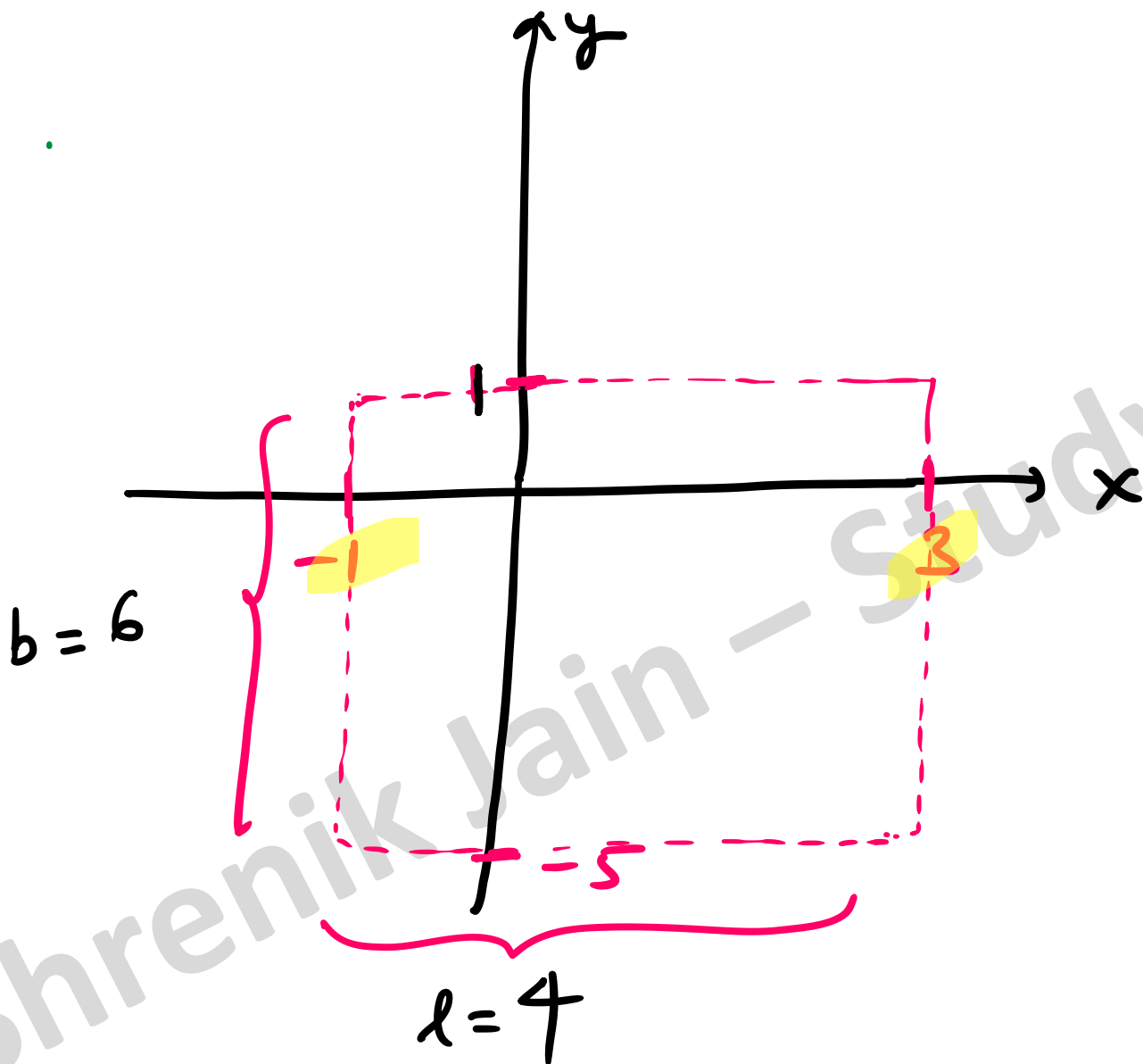
$$-(y+2) \leq 3$$

$$-y-2 \leq 3$$

$$-y \leq +5$$

$$\boxed{y \geq -5}$$

$$\boxed{-5 \leq y \leq 1}$$



$$-1 \leq x \leq 3$$

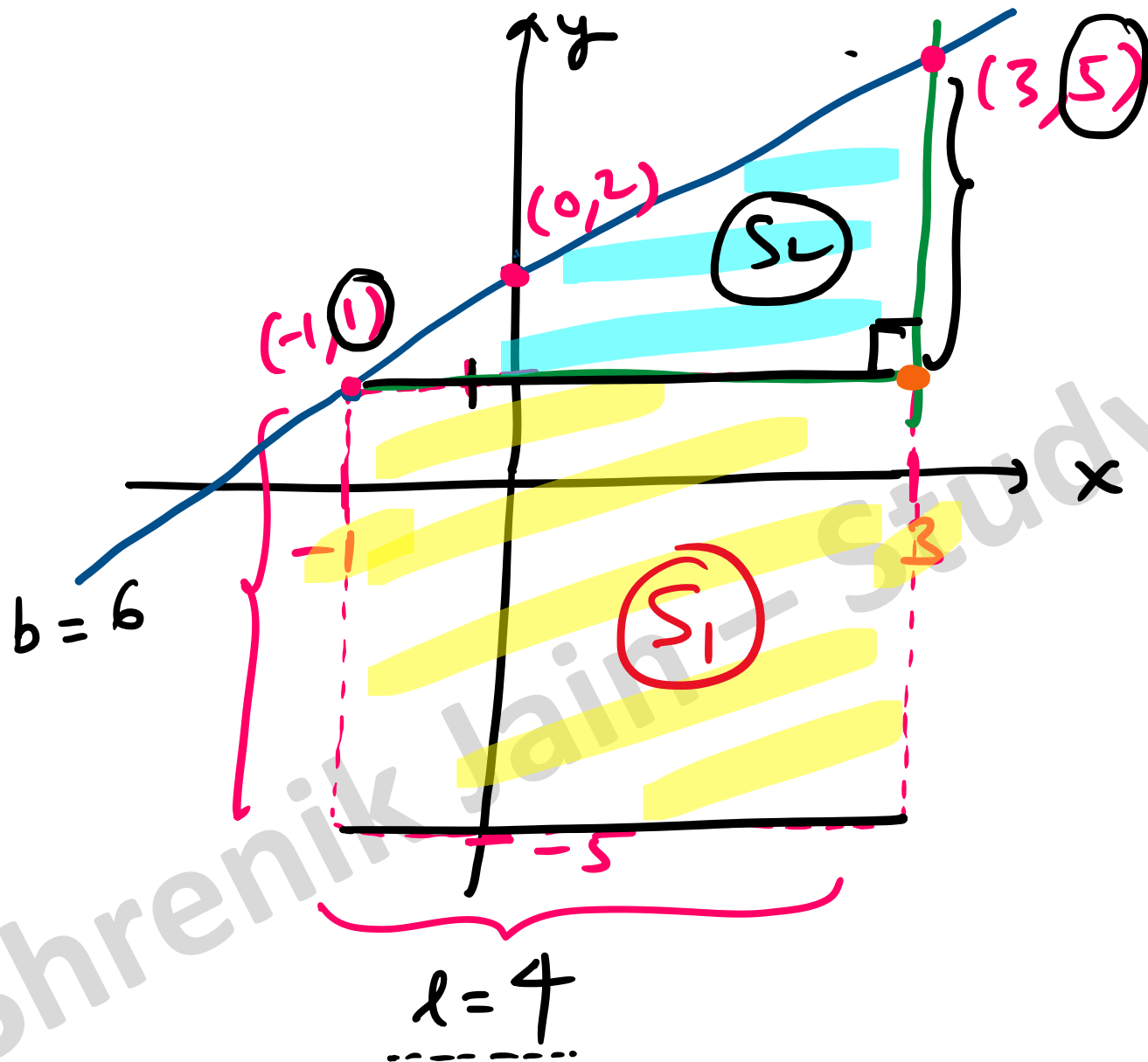
$$-5 \leq y \leq 1$$

S_1

$$A(S_1) = l b$$

$$= 4 \times 6$$

$$= 24 //$$



$$x - y \geq -2$$

$$y \geq 1$$

$$x \leq 3$$

$$x - y = -2$$

$$-1 - 1 = -2$$

$$A(S_2) = \frac{1}{2} b h$$

$$= \frac{1}{2} 4 \times 4$$

$$= 8 //$$

Q] A straight line is fit to a data set $(\ln x, y)$. This line intercepts the abscissa at $\ln x = 0.1$ and has a slope of -0.02 . What is the value of y at $x=5$ from the fit?

- (A) -0.03
- (B) -0.014
- (C) 0.014
- (D) 0.03

[GATE 2016]

①

Abscissa $\rightarrow$ x axis

Ordinate $\rightarrow$ y axis

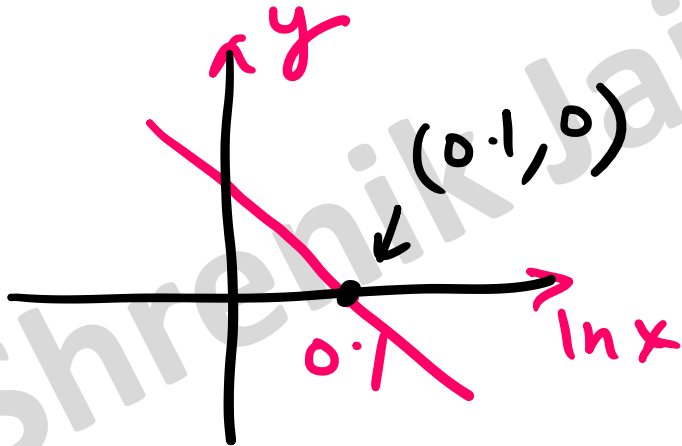
②

$$y = m\alpha + c$$

$\{x, y = \text{variable}\}$

$$y = m[\ln x] + c$$

$$y = -0.02(\ln x) + c$$



$$y = -0.02 (\ln x) + C$$

↓

$$\ln x = 0.1, y = 0$$

↓

$$0 = -0.02 (0.1) + C$$

$$C = 0.002$$

$$\therefore y = -0.02 (\ln x) + 0.002$$

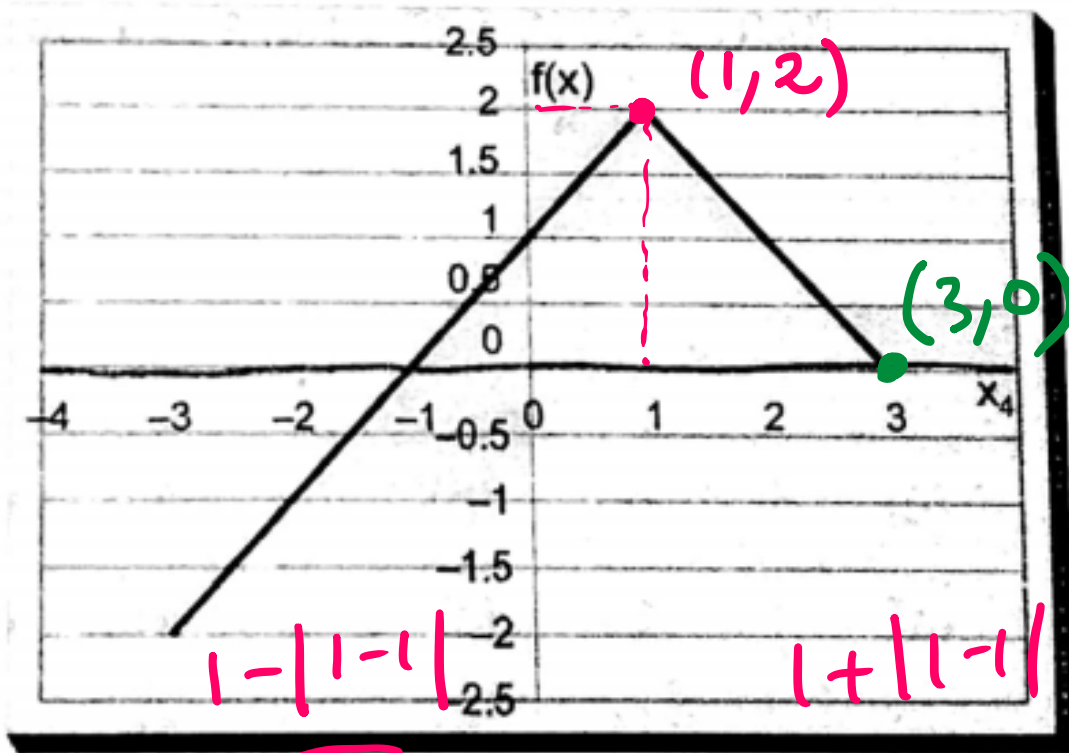
$$\therefore y = -0.02 (\ln x) + 0.002$$

$$y = -0.02 (\ln 5) + 0.002$$

$$= -0.02 [1.609] + 0.002$$

$$= -0.03 //$$

Choose the correct expression for $f(x)$ given in the graph.



$$\begin{aligned} (c) \quad f(x) &= 2 - |3 - 1| \\ &= 2 - |2| \\ &= 2 - 2 = 0 \checkmark \end{aligned}$$

$$\begin{aligned} \cancel{(d)} \quad f(x) &= 2 + |3 - 1| \\ &= 2 + |2| \\ &= 2 + 2 = 4 \neq 0 \times \end{aligned}$$

$$\cancel{(a)} \quad f(x) = 1 - |x - 1|$$

$$\cancel{(b)} \quad f(x) = 1 + |x - 1|$$

$$\checkmark (c) \quad f(x) = 2 - |x - 1|$$

$$\cancel{(d)} \quad f(x) = 2 + |x - 1|$$

[GATE-2016]

In a quadratic function, the value of the product of

the roots (α, β) is 4. Find the value of $\frac{\alpha^n + \beta^n}{\alpha^{-n} + \beta^{-n}}$

(a) n^4

☒ (b) 4^n

(c) 2^{2n-1}

(d) 4^{n-1}

[GATE-2016]

$\alpha\beta = 4$

$$= \frac{\alpha^n + \beta^n}{\frac{1}{\alpha^n} + \frac{1}{\beta^n}}$$

$$= \frac{\alpha^n + \beta^n}{\frac{\alpha^n + \beta^n}{(\alpha\beta)^n}}$$

$$= (\alpha\beta)^n$$

$$= 4^n //$$

The roots of $ax^2 + bx + c = 0$ are real and positive, a , b and c are real. Then $ax^2 + b|x| + c = 0$ has

(a) no roots

~~(b) 2 real roots~~

(c) 3 real roots

~~(d) 4 real roots~~

[GATE-2014]

M1

(i) $a=1, b=-5, c=6$

$\rightarrow x^2 - 5x + 6 = 0$

$\rightarrow (x-3)(x-2) = 0$

$\rightarrow \boxed{x = 3, 2}$

$$\rightarrow ax^2 + b|x| + c = 0$$

$$\rightarrow x^2 - 5|x| + 6 = 0$$

$$\underline{x > 0} : x^2 - 5(x) + 6 = 0$$

$$x = 2, 3$$

$$\underline{x < 0} : x^2 - 5(-x) + 6 = 0$$

$$x^2 + 5x + 6 = 0$$

$$x^2 + \underline{5}x + 6 = 0$$

$$(x + 3)(x + 2) = 0$$

$$\boxed{x = -3, -2} //$$

$\therefore$ 4 Real roots //

M2

✓ $ax^2 + bx + c = 0$

$D = b^2 - 4ac \geq 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Real & $\rightarrow D \geq 0$

$D = b^2 - 4ac$

$\rightarrow ax^2 + b|x| + c = 0$

$x > 0$

$ax^2 + bx + c = 0$

$x < 0$

$ax^2 - bx + c = 0$

$D = b^2 - 4ac$

If $|4X - 7| = 5$ then the values of $2|X| - |-X|$ is

(a) 2, $1/3$

~~(b) $1/2$~~ , 3

(c) $3/2$, 9

(d) $2/3$, 9 [GATE-2013]

M1

$$|4x - 7| = 5$$

$$\rightarrow 4x - 7 = 0 \rightarrow x = 7/4$$

$$\underline{x > 7/4} : (4x - 7) = 5 \rightarrow 4x = 12 \rightarrow x = 3 //$$

$$\underline{x < 7/4} : -(4x - 7) = 5 \rightarrow -4x + 7 = 5 \rightarrow x = 0.5 //$$

8]

$$2|x| - |-x|$$

$$\rightarrow \underline{x=3} :$$

$$2|3| - |-3|$$

$$= 2 \times 3 - (3)$$

$$= 6 - 3$$

$$= 3 //$$

$$\rightarrow \underline{x=0.5} : 2(0.5) - (0.5) = 1/2 //$$

If $|4X - 7| = 5$ then the values of $2|X| - | - X |$ is

(a) 2, $1/3$

(b) $1/2$, 3

(c) $3/2$, 9

(d) $2/3$, 9 [GATE-2013]

M2

~~A~~

$$|x| = 2 \rightarrow x = 2 \rightarrow |8 - 7| = 1 \neq 5$$

~~B~~

$$|x| = 1/2 \rightarrow x = 1/2 \rightarrow |2 - 7| = |-5| = 5$$

(C) $|x| = 3/2 \rightarrow x = 3/2 \rightarrow |6 - 7| \neq 5$

(D) $x = 2/3 \rightarrow |8/3 - 7| \neq 5$

$$2|x| - |-x| = |x|$$

Given that $f(y) = \frac{|y|}{y}$ and q is any non-zero real number, the value of $|f(q) - f(-q)|$ is

~~(a) 0~~

(b) -1

~~(c) 1~~

~~(d) 2~~

[GATE-2011]

MI

$$f(y) = \frac{|y|}{y}$$

1

, $y > 0$

-1

, $y < 0$

$$f(x) = \begin{cases} 1 & , x > 0 \\ -1 & , x < 0 \end{cases}$$

$$f(-x) = \begin{cases} 1 & , -x > 0 \\ -1 & , -x < 0 \end{cases}$$

$x < 0$
 $x > 0$

$$\therefore f(-x) = \begin{cases} -1 & , x > 0 \\ 1 & , x < 0 \end{cases}$$

$$\Rightarrow f(x) - f(-x) = \begin{cases} 2 & , x > 0 \\ -2 & , x < 0 \end{cases}$$

$$\Rightarrow |f(x) - f(-x)| = \begin{cases} |2| & , x > 0 \\ |-2| & , x < 0 \end{cases}$$

$$\therefore |f(x) - f(-x)| = 2$$

Given that $f(y) = \frac{|y|}{y}$ and q is any non-zero real number, the value of $|f(q) - f(-q)|$ is

(a) 0

(b) -1

(c) 1

~~(d) 2~~

[GATE-2011]

$q=1$

(M2)

$$|f(1) - f(-1)|$$

$$|1 - (-1)|$$

$$= |1 + 1| = 2 //$$

$$\frac{1-1}{-1} = \frac{1}{-1} = -1$$

If x is real and $|x^2 - 2x + 3| = 11$, then possible values of $|-x^3 + x^2 - x|$ include

- (a) 2, 4
- (b) 2, 14
- (c) 4, 52
- ~~(d) 14, 52~~

[EE, CS 2014, 2 Marks (Set-2)]

$$|x^2 - 2x + 3| = 11$$

$$|(x-1)^2 + 2| = 11$$

$$\Rightarrow (x-1)^2 + 2 = 11$$

$$(x-1)^2 + 2 = 11$$

$$(x-1)^2 = 9$$

$$(x-1) = \pm \sqrt{9} = \pm 3$$


$$x-1 = 3$$

$$\boxed{x = 4}$$

$$x-1 = -3$$

$$\boxed{x = -2}$$

$$\therefore |-x^3 + x^2 - x|$$

$$\underline{x = 4}$$

$$|(-4)^3 + 4^2 - 4| = 52 //$$

$$\underline{x = -2}$$

$$| -(-2)^3 + (-2)^2 - (-2) |$$

$$= 14 //$$